Epson Network Printer Technical Reference & SDK Specification Guide (2015–2026)

Edition: Print Shop & Multi-Device Network Control Architecture
Target Audience: Software Engineers, Print Shop Automation Architects, and SDK Integrators

1. Executive Summary & Architectural Scope

Modern commercial print shops, photo studios, and document centers operate heterogeneous fleets of Epson Printers connected to a single local area network (LAN/WLAN). Controlling these devices from a unified application requires a hardware capability database because each model family exposes distinct media paths, ink chemistries, page description languages (PDLs), and specialty tray geometries (such as CD/DVD platters, PVC ID card trays, rear straight-through poster feeds, and roll cutters).

While Epson has produced thousands of regional SKU numbers globally since 2015, all network-capable print shop hardware maps to **six core engineering platforms**:

1. **EcoTank Photo & Specialty Series (L800/L8000 & ET-8500/18100 Series):** 6-color dye or hybrid pigment/dye ink tank printers engineered for borderless photos, motorized CD/DVD printing, and direct PVC ID card printing.
2. **EcoTank Document & Commercial Office Series (L3000/L4000/L6000/L14000/L15000 & M-Series):** 4-color high-yield ink tank systems (including all-pigment DURABrite ET models) for high-volume A4/A3 document and duplex printing.
3. **Expression Photo & Premium Series (XP-Series):** Compact 5-color and 6-color Claria Photo HD printers featuring motorized CD/DVD trays and rear specialty media feeds.
4. **WorkForce Pro & Enterprise Series (WF-7000, WF-C5000, WF-C8000R, AM-C Series):** PrecisionCore Heat-Free business systems supporting high-speed duplexing, PCL/PostScript/IPP Everywhere, and Replaceable Ink Pack Systems (RIPS).
5. **SureColor P-Series (SC-P400 to SC-P9500):** 8-to-12-color UltraChrome pigment wide-format printers supporting fine art cotton rag, rigid posterboard (up to 1.5 mm), roll paper, and CD/DVD trays.
6. **SureColor T/F-Series & SureLab Dry Minilabs (SC-T, SC-F, SL-D Series):** CAD plotters, dye-sublimation transfer systems (for mugs, textiles, and promotional goods), and high-throughput commercial photo kiosks.

2. Network Discovery, Control Protocols & App Integration Architecture

To dynamically customize your application UI per printer (e.g., unlocking the "Print PVC ID Card" or "Print CD/DVD" module only when supported), your backend or local network daemon should use a three-layer discovery and control stack:

2.1 Protocol & Port Matrix

Protocol / API	Default Port	Supported Families	Primary Role in Print Shop Control App
mDNS / Bonjour (`_ipp._tcp`, `_pdl-datastream._tcp`)	UDP 5353	All Wi-Fi / Ethernet models (2015–2026)	Zero-configuration discovery; returns `ty` (model name), `pdl` (supported languages), `Duplex`, and `UUID`.
IPP Everywhere / AirPrint (IPP 2.0)	TCP 631	L3250+, L4260+, L6270+, L8050, L18050, L8180, L15150, WF-Pro, SC-P700/P900, SC-T	Driverless job submission (PDF, PWG-Raster, JPEG) and live capability querying via `Get-Printer-Attributes`.
Raw Socket / JetDirect (ESC/P-R, PCL, PS)	TCP 9100	All Networked EcoTank, Expression, WorkForce, SureColor	Direct byte-stream transmission of compiled Epson ESC/P-R or ESC/P2 raster jobs.
SNMP v1/v2c/v3	UDP 161	All Networked EcoTank, WorkForce, SureColor, SureLab	Real-time telemetry: per-color ink tank percentage, maintenance box capacity, page counters, and cover/jam alerts.
Epson ePOS / ESC/POS	TCP 8008 / 9100	TM-Series Receipt & CW-Series Label Printers	Receipt/ticket printing for shop billing counters (Thermal & POS Drivers).

2.2 Querying Unique Capabilities via IPP (`Get-Printer-Attributes`)

When a printer connects to the shop network, query `ipp://<printer-ip>:631/ipp/print` for the following standard attributes to auto-configure your app's print dialog:

- **`media-source-supported`:** Reveals physical input trays:
  - `'main'` / `'tray-1'`, `'tray-2'`: Front paper cassettes (e.g., EcoTank L15150, WorkForce Pro WF-7820).

    - `'rear'`: Rear vertical multi-sheet feed (supports glossy photo paper up to 300 gsm).
    - `'manual'` / `'by-pass-tray'`: Straight-through rear path for thick cardstock or posterboard up to 1.3–1.5 mm (e.g., EcoTank L8180, SureColor SC-P900).
    - `'disc'` / `'cd-tray'`: Motorized 12 cm optical disc feed for printable CDs/DVDs/Blu-rays (e.g., EcoTank L805, EcoTank L8050, EcoTank L18050, Expression Photo XP-15000).
    - `'card'` / `'id-card-tray'`: Dedicated CR80 PVC ID card tray (supported natively on EcoTank L8050 / ET-8500 / EcoTank L18050 / ET-18100).
    - `'roll'` / `'main-roll'`: Continuous roll paper feed with automatic rotary cutter (e.g., SureColor SC-T3100, SureColor SC-F500, SureLab SL-D800).
- **`media-col-database` & Margins:** Returns supported dimensions and whether `media-bottom-margin`, `media-top-margin`, `media-left-margin`, and `media-right-margin` can be set to `0` (Borderless) or require `300` (3.0 mm standard margin).
- **`sides-supported`:** Returns `'one-sided'` only or `'two-sided-long-edge'` / `'two-sided-short-edge'` for auto-duplex models.

2.3 Specialty Media Print Geometry (CD/DVD & PVC ID Cards)

For print shops offering optical disc and plastic ID card printing, your app must enforce strict coordinate clipping before rasterizing jobs:

- **CD/DVD/Blu-ray Optical Discs:**
    - **Outer Printable Diameter:** Standard 116 mm (adjustable up to 118 mm or down to 70 mm for 8 cm mini-discs).
    - **Inner Hub Hole Diameter:** Standard 43 mm (inkjet hub printable discs allow down to 18 mm inner diameter).
    - **App Requirement:** Apply a circular alpha mask excluding pixels `< 18 mm` and `> 118 mm` from the center to prevent ink pooling on the plastic tray or optical data tracks.
- **PVC Plastic ID Cards (CR80 Standard):**
    - **Card Dimensions:** 85.60 × 53.98 mm (3.375 × 2.125 in), thickness 0.76 mm (30 mil), ISO/IEC 7810 ID-1 standard.
    - **Tray Slot Geometry:** On the EcoTank L8050 and EcoTank L18050, the multi-purpose tray holds **2 PVC cards simultaneously** (Slots 1 and 2) with 1–2 mm full-bleed overprint grooves. Note: Must use **Inkjet-Receptive Coated PVC Cards** (standard un-coated thermal PVC cards will not absorb aqueous dye/pigment ink).

3. Master Specification Tables by Series (2015–2026)

3.1 EcoTank Photo & Specialty Series (6-Color Photo, CD/DVD & PVC ID Card)

These are the primary revenue-generating specialty printers in retail photo and print shops, offering ultra-low-cost 6-color printing and direct rigid tray feeding.

Model (Global / Regional SKUs)	Release Window	Max Format & Borderless	Printhead, Resolution & Droplet	Ink System & Channels	Print Speed (Draft / ISO / 4×6 Photo)	Paper Trays, Grammage & Thickness	Unique Shop Capabilities & Specialties
EcoTank L805 (L810 / L850 MFP variants)	2015–2022	A4 / Letter (Borderless up to...	Micro Piezo (90×6 nozzles), ...	6-Color Dye (T673: K, C, M, ...	37 ppm Draft / 5.1 ipm ISO / 1...	Rear Feed (120 sheets plain /...	Motorized **CD/DVD tray**; aftermarket 2-card PVC tray compatible via CD slot; Wi-Fi (L805) or SD/LCD (L810/L850).
EcoTank L1800	2015–2023	A3+ / 13×19" (Borderless up ...	Micro Piezo (90×6 nozzles), ...	6-Color Dye (T673: K, C, M, ...	15 ppm Draft / 2.6 ipm ISO / ...	Rear Feed (100 sheets plain ...	Industry workhorse for A3+ borderless photos, film positives, and DTF/transfer conversions; USB 2.0 (requires print server on LAN).
EcoTank L8050 (ET-18100 A4 sibling)	2023–2026	A4 / Letter (Borderless up to...	Micro Piezo Heat-Free (180×...	6-Color Dye (057 / 108: K, C,...	22 ppm Draft / 8.0 ipm ISO / ...	Rear Feed (100 sheets); 64–...	Native **PVC ID Card tray (2 cards)** + **CD/DVD tray** (Guide); user-replaceable maintenance box (C9345) & pickup rollers; Wi-Fi + Smart Panel.
EcoTank L18050 (ET-18100)	2023–2026	A3+ / 13×19" (Borderless up ...	Micro Piezo Heat-Free (180×...	6-Color Dye (057 / 108: K, C,...	22 ppm Draft / 8.0 ipm ISO / ...	Rear Feed (100 sheets); 64–...	Native **A3+ borderless**, **PVC ID Card tray (ET-18100 Video)**, and **CD/DVD printing**; replaceable C9345 waste ink box; Wi-Fi & Wi-Fi Direct.
EcoTank L8160 / ET-8500	2021–2026	A4 / Letter (Borderless up to...	Micro Piezo (360×2 Black, 1...	6-Color Claria ET Premium (...	32 ppm Draft / 16 ipm B&W, 1...	5 Paper Paths: Front Cassett...	Hybrid **Pigment Black (sharp text) + Dye Photo Black & Grey** (B&W photo depth); **1.3 mm rigid media pass**; Auto-Duplex; **CD/DVD tray**; Ethernet + Wi-Fi + Scan/Copy.
EcoTank L8180 / ET-8550	2021–2026	A3+ / 13×19" (Borderless up ...	Micro Piezo (360×2 Black, 1...	6-Color Claria ET Premium (...	32 ppm Draft / 16 ipm B&W, 1...	5 Paper Paths: Front Cassett...	Flagship all-in-one shop photo printer: **A3+ borderless**, **1.3 mm rigid posterboard pass**, **CD/DVD**

Model (Global / Regional SKUs)	Release Window	Max Format & Borderless	Printhead, Resolution & Droplet	Ink System & Channels	Print Speed (Draft / ISO / 4×6 Photo)	Paper Trays, Grammage & Thickness	Unique Shop Capabilities & Specialties
							printing, A4 auto-duplex, Ethernet + Wi-Fi + AirPrint.

3.2 EcoTank Document, Office & Commercial Series (A4 & A3 Network MFPs)

Used in print shops for high-volume document printing, book/manual binding, scanning, and A3 color posters/flyers.

Model (Global / Regional SKUs)	Release Window	Max Format & Duplex	Printhead, Resolution & Droplet	Ink System & Chemistry	Print Speed (ISO / Draft)	Paper Trays, ADF & Grammage	Unique Shop Capabilities & Specialties
EcoTank L3250 (L3150 / L3550 / ET-2720 / ET-2850)	2018–2026	A4 / Letter (Manual Duplex; Auto on L3550/ET-2850)	Micro Piezo (180 BK, 59×3 C...	4-Color Dye (003 / 103 / 52...	10 ipm B&W / 5.0 ipm Color (33 ppm Draft)	Rear Feed (100 sheets); 64–300 gsm; Borderless up to 4R	Entry-level network shop printer; Wi-Fi + Wi-Fi Direct; L3550 update adds 20 ppm head & replaceable C9344 maintenance box.
EcoTank L4260 (L4160 / L4360 / ET-2760 / ET-2850)	2018–2026	A4 / Letter (**Auto-Duplex** A4)	Micro Piezo (180 BK, 59×3 C...	4-Color Hybrid (001 / 101 / 5...	10.5 ipm B&W / 5.0 ipm Color (33 ppm Draft)	Rear Feed (100 sheets); 64–300 gsm; Borderless up to A4	Water-resistant Pigment Black for sharp document text + Dye CMY for borderless A4 photos; Auto-Duplex; Wi-Fi Direct + AirPrint.
EcoTank L6270 / **L6290** (L6170 / L6190 / ET-3850 / ET-4850)	2018–2026	A4 / Letter (**Auto-Duplex** Print + 30-sheet ADF)	PrecisionCore Heat-Free (40...	4-Color Hybrid (001 / 101 / 5...	15.5 ipm B&W / 8.5 ipm Color (33 ppm Draft)	Front Cassette (250 sheets); 64–250 gsm; Borderless up to A4	Built for workgroup shops: **Ethernet + Wi-Fi**, 30-sheet ADF (Scan/Copy; Fax on L6290), user-replaceable C13T04D100 maintenance box.
EcoTank L6490 (L6460 / ET-5150 / ET-5170)	2021–2026	A4 / Letter (**Auto-Duplex** Print + 35-sheet ADF)	PrecisionCore Heat-Free (40...	4-Color All-Pigment DURABr...	17 ipm B&W / 9.5 ipm Color (37 ppm Draft)	Front Cassette (250 sheets) + Rear Single-Sheet Feed; 64–256 gsm	100% water/highlighter-resistant **All-Pigment** prints on both sides; Ethernet + Wi-Fi + AirPrint.
EcoTank L6570 / L6580 (ET-5850 / ET-5880)	2020–2026	A4 / Letter (**Auto-Duplex** Print & Single-Pass Duplex ADF)	PrecisionCore 4S Head (800...	4-Color All-Pigment DURABr...	**25 ipm B&W / 25 ipm Color** (32 ppm Draft; 4.5s FPOT)	2×250-sheet Front Trays + 50-sheet Rear Feed (550 total); 64–256 gsm	High-speed commercial A4 workhorse; 50-sheet **Single-Pass Duplex ADF** (27.5 ipm duplex scan); L6580/ET-5880 adds **PCL5/PCL6 & PostScript 3**.

Model (Global / Regional SKUs)	Release Window	Max Format & Duplex	Printhead, Resolution & Droplet	Ink System & Chemistry	Print Speed (ISO / Draft)	Paper Trays, ADF & Grammage	Unique Shop Capabilities & Specialties
EcoTank L14150 (ET-15000)	2020–2026	**A3+ Print** (via Rear Feed) / A4 Auto-Duplex & A4 Scan/ADF	PrecisionCore Heat-Free (40...	4-Color Hybrid (001 / 101 / 5...	17 ipm B&W / 9.0 ipm Color (A4); 3.5 ipm A3	Front Cassette (250 sheets A4) + Rear Feed (20 sheets up to **A3+/13×19"**); 64–255 gsm	Compact A4 footprint that prints **A3+ via rear feed**; 35-sheet A4 ADF; Ethernet + Wi-Fi + AirPrint; C13T04D100 maintenance box.
EcoTank L15150 / L15160 / L15180 (ET-16600 / ET-16650 / ET-16680)	2020–2026	**A3+ Print** / **A3 Auto-Duplex Print, Scan, Copy & Fax**	PrecisionCore 4S Head (800...	4-Color All-Pigment DURABr...	**25 ipm B&W / 25 ipm Color** (L15160/15180; L15150 is 25/12 ipm)	2×250-sheet **A3 Front Trays** + 50-sheet Rear A3+ Feed; 50-sheet **A3 Duplex ADF**; 64–255 gsm	Full **A3 Auto-Duplex** print & scan; Barcode Mode for crisp retail labels; **L15180 / ET-16680** adds native **PCL5c/PCL6 & PostScript 3**.
EcoTank Monochrome M1120 / M2140 / M3170 / M15180	2019–2026	A4 (M1120/2140/3170) / **A3 (M15180)**; Auto-Duplex on M2140+	PrecisionCore Heat-Free (40...	1-Color Pigment Black (005 /...	15 ipm (M1120) / 20 ipm (M2140/3170) / **25 ipm (M15180)**	250-sheet Cassette + Rear Feed (550 sheets A3 on M15180); 64–256 gsm	Laser-replacement monochrome tank printers for invoice, receipt, and textbook printing at ultra-low CPP and < 20W power draw.

3.3 Expression Photo & Premium Series (Claria Photo HD & CD/DVD Specialty)

Popular in boutique photo studios and European/North American print shops for 6-color Claria Photo HD prints and CD/DVD disc customization.

Model	Release Window	Max Format & Borderless	Printhead, Resolution & Droplet	Ink System & Channels	Print Speed (ISO / 4×6 Photo)	Paper Trays & Grammage	Unique Shop Capabilities & Specialties
Expression Photo HD XP-15000	2018–2026	**A3+ / 13×19"** (Borderless up to A3+); **A4 Auto-Duplex**	Micro Piezo (180×6 nozzles)...	6-Color Claria Photo HD (312/314 or 378/478: K, C, M, Y, **Red, Grey**)	9.2 ipm B&W / 9.0 ipm Color / 27 sec (4R Photo)	Front Cassette (200 sheets A4, Auto-Duplex) + Rear Specialty Feed (50 sheets up to **A3+ & 23 mil / 0.6 mm**); CD Tray	Red & Grey inks replacing LC/LM for wider gamut and neutral B&W; motorized **CD/DVD tray**; Ethernet + Wi-Fi + AirPrint.
Expression Photo XP-8700 (XP-8500 / XP-8600)	2017–2026	A4 / Letter (Borderless up to A4); **Auto-Duplex**	Micro Piezo (180×6 nozzles)...	6-Color Claria Photo HD (312 / 378: K, C, M, Y, LC, LM)	9.5 ipm B&W / 9.0 ipm Color / 10 sec (4R Photo)	Dual Front Cassettes (100 A4 + 20 Photo) + Rear Specialty Pass (up to 0.6 mm card); CD Tray	Motorized output tray + **CD/DVD printing**; rear thick media feed; 4.3" touchscreen; Wi-Fi Direct + AirPrint.
Expression Photo XP-970 (XP-960)	2016–2024	**A3** (via Rear Single-Sheet Feed) / A4 Auto-Duplex + Scan/Copy	Micro Piezo (180×6 nozzles)...	6-Color Claria Photo HD (277 / 24: K, C, M, Y, LC, LM)	8.5 ipm B&W / 8.0 ipm Color / 11 sec (4R Photo)	Dual Front Cassettes + Rear Single-Sheet A3 Feed; CD Tray; 64–300 gsm	Ultra-compact A4 MFP footprint capable of printing **borderless A3 photos** via rear slot + **CD/DVD printing** + Ethernet/Wi-Fi.

Model	Release Window	Max Format & Borderless	Printhead, Resolution & Droplet	Ink System & Channels	Print Speed (ISO / 4×6 Photo)	Paper Trays & Grammage	Unique Shop Capabilities & Specialties
Expression Premium XP-6100 / XP-7100 (XP-6000)	2018–2025	A4 / Letter (Borderless up to A4); **Auto-Duplex** (ADF on XP-7100)	Micro Piezo (360 Pigment B…	5-Color Claria Premium (302 / 202: **Pigment BK** + Dye Photo BK, C, M, Y)	15.8 ipm B&W / 11.3 ipm Color / 15 sec (4R Photo)	Dual Front Cassettes (100 A4 + 20 Photo) + **CD/DVD Tray**; 64–300 gsm	Combines Pigment Black for crisp document text with 4 dye inks for glossy photos + dedicated **CD/DVD tray**.

3.4 WorkForce Pro & Enterprise Series (High-Speed Commercial Document & SRA3/A3+ Production)

Engineered for high-volume copy shops and commercial print centers requiring fast First Page Out Time (FPOT), heavy duty cycles, and enterprise PDL support (PCL6, PostScript 3, IPP Everywhere).

Model	Release Window	Max Format & Duplex	Printhead & Resolution	Ink System & Yield	Print Speed (ISO / Duplex)	Paper Trays, ADF & Grammage	Unique Shop Capabilities & Specialties
WorkForce Pro WF-7820 / **WF-7840** (WF-7710 / WF-7720 / WF-7310)	2017–2026	**A3+ / 13×19" Print** (Borderless up to A3+); **A3 Auto-Duplex Print, Scan, Copy, Fax**	PrecisionCore 4S (800×4 nozzles), 4800×2400 dpi, 3.8 pl	4-Color DURABrite Ultra **All-Pigment** (405 / 812: K, C, M, Y)	**25 ipm B&W / 12 ipm Color** (16 / 9 ipm Duplex)	1×250 (WF-7820) or 2×250 (WF-7840) A3 Front Cassettes + 50-sheet Rear A3+ Feed; 50-sheet A3 Duplex ADF; 64–255 gsm	Low upfront cost A3+ shop workhorse; prints envelopes, 255 gsm cardstock, and borderless A3+ flyers; Ethernet + Wi-Fi + ESC/P-R.
WorkForce Pro WF-C5790 / **WF-C5890** (WF-C5290 / WF-C5390 single-function)	2018–2026	A4 / Legal (**Auto-Duplex** Print & Single-Pass Duplex ADF)	PrecisionCore 4S (800×4 nozzles), 4800×1200 dpi, 3.8 pl	4-Color DURABrite Ultra **RIPS Ink Packs** (T94 / T11 up to 10,000 BK / 5,000 CMY yield)	**24–25 ipm B&W & Color** (15–16 ipm Duplex; 4.8s FPOT)	250-sheet Cassette + 80-sheet Rear MP Tray (expandable to 1,330 sheets); 64–256 gsm	**Native PCL5c, PCL6, PostScript 3, PDF Direct & ESC/P-R**; 50-sheet single-pass duplex ADF; high-yield foil ink packs located in bottom drawer.
WorkForce Pro WF-C878R / **WF-C879R** (WF-C869R / EM-C8100R)	2018–2026	**A3+ / 13×19" Print**; A3 Auto-Duplex + Single-Pass A3 ADF	PrecisionCore 4S (800×4 nozzles), 4800×1200 dpi, 3.7 pl	4-Color DURABrite Pro High-Capacity **RIPS Bags** (up to **86,000 BK / 50,000 CMY** pages)	**25–26 ipm B&W / 24–25 ipm Color** (18 ipm Duplex; 5.5s FPOT)	250-sheet Cassette + 85-sheet Rear MP Tray (expandable to **1,835 sheets**); 64–300 gsm	Ultra-high-yield RIPS bags eliminate frequent refills in busy copy shops; **PCL6, PostScript 3, IPP Everywhere**; optional staple finisher.
WorkForce Enterprise AM-C5000 (AM-C4000 / AM-C6000 / WF-C20600 / C21000)	2020–2026	**A3 / SRA3** (Full-bleed A3 via SRA3 crop); Auto-Duplex	PrecisionCore **Linehead** (fixed page-wide array, 600×2400 dpi)	4-Color DURABrite Pro Pigment Cartridges (50,000 BK / 30,000 CMY)	**40 / 50 / 60 ppm** (AM-C4000/5000/6000); up to **100 ppm** (WF-C21000 / AM-C10000)	2×500-sheet Cassettes + 150-sheet MP Tray (up to 5,150 sheets); **52–300 gsm**; 150-sheet 120 ipm Duplex ADF	Fixed **Linehead** technology prints at 40–100 ppm with < 120W power draw; supports **booklet maker, saddle-stitching, hole punching, and stapling** via IPP `finishings`.

3.5 SureColor P-Series (Professional Photo, Fine Art, Rigid Posterboard & CD/DVD)

Used in pro photo labs, framing shops, and graphic design studios for archival gallery prints, canvas, thick matboard, and panoramic banners.

Model	Release Window	Max Format & Roll/Sheet	Printhead & Resolution	Ink System & Channels	Media Paths, Thickness & Specialties	Network & Control Languages
SureColor SC-P400 / SC-P600	2015–2020	**A3+ / 13"** (Sheet + 13" Roll Holder + CD Tray)	Micro Piezo AMC (180×8 nozzles), 5760×1440 dpi, 1.5–2 pl	**8-Color UltraChrome HG2** (SC-P400, w/ Red, Orange, Gloss Optimizer) or **9-Color UltraChrome HD** (SC-P600)	ASF Sheet Feeder + **13" Roll Paper** + Fine Art Path + Straight Rigid Path (**up to 1.3 mm posterboard**) + **CD/DVD Tray**	Ethernet, Wi-Fi, USB; ESC/P3 / ESC/P-R.
SureColor SC-P700 (13" A3+) / **SC-P900** (17" A2)	2020–2026	**13" A3+ (SC-P700)** / **17" A2 (SC-P900)**; Sheet + Roll + CD	Micro Piezo AMC F10 (180×10 nozzles), 5760×1440 dpi, 1.5 pl	**10-Color UltraChrome PRO10** (PK, MK, C, LC, VM, VLM, Y, GY, LGY, **Violet**) – **No PK/MK switching**	Dedicated channels for both Photo Black & Matte Black (zero ink waste switching); **Built-in 13" Roll Unit** (SC-P700; optional 17" Roll Bundle on SC-P900); **1.5 mm rigid board** pass; **CD/DVD tray**; Black Enhance Overcoat (BEO).	Ethernet, Wi-Fi (2.4/5 GHz), AirPrint, IPP Everywhere, ESC/P3, Epson Print Layout.
SureColor SC-P800 (17" A2) / **SC-P5000** (17" Proofing)	2015–2025	**17" / A2+** (Sheet + 17" Roll Adapter)	Micro Piezo TFP / AMC, 2880×1440 dpi, 3.5 pl	**9-Color UltraChrome HD** (SC-P800) or **11-Color UltraChrome HDX** (SC-P5000 w/ Orange, Green, Violet)	Top Sheet Feeder + Front Straight Rigid Board Path (**up to 1.5 mm**) + 17" Roll Unit (2" or 3" core); SC-P5000 covers 99% Pantone Solid Coated.	Ethernet 1000Base-T, Wi-Fi (SC-P800), ESC/P3, SpectroProofer option.
SureColor SC-P6000 / P7000 / P8000 / P9000 & **SC-P7500 / P9500**	2015–2026	**24" / A1+** (P6000/7000/7500) & **44" / B0+** (P8000/9000/9500)	PrecisionCore TFP / µTFP12 (300×12 nozzles), 2400×1200 dpi, 3.5 pl	**8-to-12-Color UltraChrome HDX / PRO12** (350 ml / 700 ml cartridges)	Roll Paper (2"/3" core, auto rotary cutter), Cut Sheet, and Rigid Posterboard (**up to 1.5 mm**); borderless roll printing on 10"–44" widths.	Gigabit Ethernet, Optional Adobe PostScript 3 / Internal HDD Server, ESC/P3.

3.6 SureColor T-Series (CAD/GIS), F-Series (Dye-Sublimation) & SureLab (Dry Photo Minilabs)

Commercial production machines commonly networked into multi-service print shops for architectural blueprints, custom merchandise (mugs, T-shirts, lanyards), and high-speed event/studio photo fulfillment.

Model	Category & Release	Max Width & Media Handling	Printhead, Ink & Resolution	Speed & Throughput	Unique Shop Capabilities & App Integration Notes
SureColor SC-T3100 (24" A1) / SC-T5100 (36" A0) & SC-T3400 / T5400	Technical CAD / Blueprint / Poster Plotter (2018–2026)	**24" (610 mm) or 36" (914 mm) Roll** + **50-sheet A4/A3 Auto Sheet Feeder** + Cut Sheets up to A1/A0; 0.05–0.27 mm	PrecisionCore MicroTFP, 2400×1200 dpi, 4.0 pl; **4-Color UltraChrome XD2 All-Pigment** (bottles on T3100x, cartridges on T3100/5100)	**31–34 sec per A1** plot (22–25 sec on T3400/5400)	**Automatic Roll-to-Cut-Sheet Switching** (holds both a 24"/36" roll and 50 sheets of A4/A3 in the ASF simultaneously—ideal for mixed network jobs); supports **HP-GL/2, RTL, ESC/P-R, AirPrint, Wi-Fi Direct, Gigabit LAN**.
SureColor SC-F100 / SC-F170	A4 Desktop Dye-Sublimation Transfer Printer (2020–2026)	A4 / Legal Front Cassette (150 sheets); 64–90 gsm transfer paper	PrecisionCore MicroTFP (400 BK, 128×3 CMY), 1200×600 dpi; **4-Color UltraChrome DS Sublimation Ink** (140 ml bottles)	~65 sec per A4 transfer sheet	Purpose-built for print shops producing **custom mugs, phone cases, mousepads, caps, and lanyards**; driver & firmware natively support **Auto Horizontal Mirror/Flip** and DS transfer ICC profiles over Ethernet/Wi-Fi.
SureColor SC-F500 / **SC-F570**	24" Dye-Sublimation Roll & Sheet Printer (2019–2026)	**24" (610 mm) Roll** + **50-sheet A4/A3 Auto Sheet Feeder** + Cut Sheets up to 24"; Auto Rotary Cutter	PrecisionCore MicroTFP, 2400×1200 dpi; **4-Color UltraChrome DS** (140 ml refillable bottles, refillable while printing)	**62 sec per A1** transfer sheet	Seamless switching between 24" rolls (for jerseys, cushions, banners) and A4/A3 sheets; Wi-Fi, Ethernet, USB; anti-dust design with Nozzle Verification Technology (NVT).
SureLab SL-D700 / SL-D800	6-Color Commercial Roll Dry Photo Minilab (2015–2026)	**Roll Widths:** 89 mm (3.5"), 102 mm (4"), 127 mm (5"), 152 mm (6"), 203 mm (8"), **210 mm (A4)**; Print lengths **50 mm to 1000 mm (1m Panorama)**	Micro Piezo, 1440×720 dpi, 2.5 pl; **6-Color UltraChrome D6-S / D6r-S** (200 ml cartridges: K, C, M, Y, LC, LM)	**360 prints/hr (4×6" Standard)** / **430 prints/hr (High Speed)**; 17 sec single Quick Print	Built-in heavy-duty rotary cutter cuts roll into exact photo lengths (4×6", 5×7", 8×10", 8×39" panoramas) on Glossy, Lustre, or Matte roll media; rated for **200,000 prints**.
SureLab SL-D1000 / SL-D1000A	Duplex Commercial Roll & Sheet Dry Minilab (2021–2026)	**Roll (89–210 mm width)** + **Auto Duplex Cut-Sheet Feeder** (SL-D1000A includes 100-sheet Duplex Cassette for photobooks/calendars)	Micro Piezo Heat-Free, 1440×720 dpi; **6-Color UltraChrome D6r-S** (**250 ml RIPS Ink Packs** – 25% more ink, less waste)	**385 prints/hr (4×6" Standard)** / **460 prints/hr (Max)**	Adds **Native Wi-Fi & Gigabit Ethernet**, **Auto-Duplex sheet printing** (for double-sided greeting cards, calendars, and photobooks), nozzle detection, and 400,000-print lifespan.
SureLab SL-D500	Stackable Simplex/Duplex 4×6" Photo Production Printer (2022–2026)	**100-sheet Auto Sheet Feeder**; 9×13 cm and **10×15 cm (4×6")**	Micro Piezo Heat-Free, 1440×720 dpi; **6-Color UltraChrome D6r-S** (70 ml integrated CISS ink tanks)	**180 prints/hr (Standard)** / **250 prints/hr (Max)** per unit	Ultra-compact (7 kg, 0.12 m² footprint); **up to 3 units stack vertically** into a single network

Model	Category & Release	Max Width & Media Handling	Printhead, Ink & Resolution	Speed & Throughput	Unique Shop Capabilities & App Integration Notes
		Borderless Simplex & Auto-Duplex			production tower (up to 750 prints/hr combined) via **Ethernet & Wi-Fi**.
ColorWorks CW-C8000 (CW-C6000 / CW-C6500 / C7500)	Industrial On-Demand Color Label Printer (2020–2026)	Roll or Rear Fanfold Labels (**25.4 mm to 108 mm / 4.25" width**; up to 215.9 mm / 8.5" on C6500); Built-in Auto Cutter or Peeler	PrecisionCore Linehead (CW-C8000/C7500) or Serial Head (C6000/C6500), 1200×600 dpi; **4-Color UltraChrome DL Pigment** (High-capacity ink packs on C8000)	Up to **300 mm/sec (11.8 ips)**	Prints full-color waterproof/GHS product labels on Glossy, Matte, BOPP, and Synthetic die-cut rolls; native **ESC/Label, ZPL II emulation, SAP, Ethernet, and Wi-Fi**.

4. App Developer Implementation Guide: JSON Capability Schema & SNMP Telemetry

4.1 Recommended Per-Printer JSON Capability Profile

Store or dynamically generate the following JSON schema per discovered printer on the shop network so your frontend UI automatically adapts its paper size dropdowns, specialty templates (CD/DVD, ID Card, Mug Sublimation), and tray selectors:{


```
 "model_id": "EPSON_L18050",

 "series": "EcoTank Photo Specialty",

 "network": {

  "ip": "192.168.1.142",

  "protocols": ["IPP", "RAW_9100", "SNMP_V2C", "MDNS"],

  "pdl_supported": ["ESC/P-R", "PWG-Raster", "AppleRaster", "JPEG"]

 },

 "printhead": {

  "technology": "Micro Piezo Heat-Free",

  "max_dpi": [5760, 1440],
```

```
  "min_droplet_pl": 1.5,
  "channels": ["K", "C", "M", "Y", "LC", "LM"],
  "ink_chemistry": "DYE_6C",
  "sublimation_mirror_default": false
},
"capabilities": {
  "max_format": "A3+",
  "max_dimensions_mm": [329, 483],
  "max_banner_length_mm": 2000,
  "auto_duplex": false,
  "borderless_sizes": ["4×6", "5×7", "8×10", "A4", "A3", "A3+"],
  "specialty_trays": {
    "cd_dvd_disc": {
      "supported": true,
      "ipp_media_source": "disc",
      "outer_diameter_mm": 116,
      "inner_diameter_mm": 18
    },
    "pvc_id_card": {
      "supported": true,
      "ipp_media_source": "card",
```

```
      "slots": 2,
      "card_size_mm": [85.6, 53.98],
      "thickness_mm": 0.76
    },
    "rigid_board_pass": {
      "supported": false,
      "max_thickness_mm": 0.3
    },
    "roll_feeder": {
      "supported": false,
      "auto_cutter": false
    }
  }
 }
}
```


4.2 SNMP Telemetry OIDs for Live Fleet Monitoring

Your shop control dashboard can poll every networked Epson printer via SNMP v1/v2c (UDP Port 161, Community: public) using standard Printer-MIB (RFC 3805) and Epson Private Enterprise OIDs (1.3.6.1.4.1.1248) to display real-time ink levels and alert operators before a job fails mid-print:

- **Device Model Description (hrDeviceDescr):** `1.3.6.1.2.1.25.3.2.1.3.1` (Returns e.g. `"EPSON L18050 Series"` or `"EPSON WF-C5890 Series"`).
- **Serial Number (prtGeneralSerialNumber):** `1.3.6.1.2.1.43.5.1.1.17.1`
- **Total Lifetime Page Counter (prtMarkerLifeCount):** `1.3.6.1.2.1.43.10.2.1.4.1.1`

- **Consumable / Ink Channel Names (prtMarkerSuppliesDescription):** `1.3.6.1.2.1.43.11.1.1.6.1.<index>` (Enumerates Black, Cyan, Magenta, Yellow, Light Cyan, Light Magenta, and Maintenance Box).
- **Consumable Max Capacity (prtMarkerSuppliesMaxCapacity):** `1.3.6.1.2.1.43.11.1.1.8.1.<index>`
- **Consumable Current Level (prtMarkerSuppliesLevel):** `1.3.6.1.2.1.43.11.1.1.9.1.<index>` (Divide by Max Capacity × 100 for exact tank/cartridge percentage; Maintenance Box level alerts shop owners when C9344/C9345/T04D1 waste pad needs replacement).
- **Active Printer Alert Codes (prtAlertCode & prtAlertDescription):** `1.3.6.1.2.1.43.18.1.1.7.1` and `1.3.6.1.2.1.43.18.1.1.8.1` (Detects Paper Out, Paper Jam, Cover Open, or CD/ID Card Tray Misaligned).

4.3 Job Routing Logic for Multi-Printer Shop Queues

When a shop operator submits a job in your app, apply the following automatic routing rules across your connected printers:

1. **PVC ID Cards (CR80 85.6×54 mm):** Route exclusively to printers where `specialty_trays.pvc_id_card.supported == true` (EcoTank L8050, EcoTank L18050, or L805/L850 configured with an aftermarket CD-slot card tray). Prompt the operator to load 1 or 2 inkjet-coated PVC cards into the front tray.
2. **CD / DVD / Blu-ray Disc Face Printing:** Route to any 6-color photo model with `specialty_trays.cd_dvd_disc.supported == true` (EcoTank L805, L8050, L18050, L8160/L8180, XP-8700, XP-15000, or SureColor SC-P700/P900).
3. **Rigid Posterboard / Thick Invitation Stock (0.6 mm – 1.5 mm):** Route to printers with a straight-through rear/front pass (EcoTank L8160/L8180 up to 1.3 mm; SureColor SC-P700/P900 up to 1.5 mm).
4. **High-Volume Double-Sided Documents & Booklets (A4/A3):** Route to PrecisionCore all-pigment devices (EcoTank L6570, L15150/L15180, WorkForce Pro WF-C5890, WF-C878R, or WorkForce Enterprise AM-C5000) to prevent bleed-through and curling.
5. **High-Throughput 4×6" / 5×7" / Panoramic Event Orders:** Route to SureLab SL-D800, SL-D1000A, or a stacked tower of SureLab SL-D500 units.
6. **Sublimation Merchandise (Mugs, T-Shirts, Lanyards):** Route to SureColor SC-F100 / SC-F170 (A4) or SC-F500 (24" Roll) with automatic horizontal image mirroring enabled.